STUDY MODULE 06

Classical Machine Learning: A Deep Dive

History, Definition, and Core Principles

Before deep neural networks and modern generative AI captured global headlines, the foundation of machine learning was built on elegant mathematical and statistical techniques. This approach is known today as **Classical Machine Learning**.

1. What Is Classical Machine Learning?

At its heart, classical machine learning relies on algorithms that take numerical data, find mathematical relationships, and make predictions. Unlike modern deep learning (which figures out everything automatically), classical ML requires human engineers to perform *feature engineering*—meaning humans must pick and choose which specific pieces of data matter most.

> **The Real-World Analogy:** If you want a classical computer to predict house prices, a human expert must first tell the computer what features to look at (e.g., square footage, number of bedrooms, and age of the house). The algorithm then calculates how much weight to give each feature.

2. A Brief History of Classical ML

Machine learning didn't appear overnight. Its roots stretch back decades through major milestones:

- **1950s (The Spark):** Arthur Samuel at IBM coins the term "machine learning" in 1959 while creating a computer program that could play checkers better than its creator.
- **1957 (The Perceptron):** Frank Rosenblatt invents the Perceptron, the earliest mathematical model of a single artificial neuron, laying the groundwork for pattern recognition.
- **1960s–1970s (Statistical Methods):** Researchers heavily develop linear regression, Bayesian classifiers, and clustering algorithms based on solid mathematical probability.
- **1980s–1990s (The Rise of Practical Algorithms):** Decision trees, support vector machines (SVMs), and random forests emerge as practical tools used by early commercial software.

3. Key Pillars of Classical ML

Classical machine learning models remain popular and widely used today because they possess distinct advantages over massive deep learning models:

- **High Interpretability:** Unlike "black box" neural networks, classical algorithms like decision trees allow humans to see exactly why a decision was made step-by-step.
- **Low Data Requirements:** They can learn effectively from hundreds or thousands of examples, rather than requiring millions of data points.
- **Fast Training Speeds:** They require minimal computing power and can run easily on everyday laptops or small embedded devices.

4. Core Classical Algorithms at a Glance

Algorithm Family	How It Works	Common Use Case
Linear Models	Fits a straight line or threshold to data points to predict continuous values or categories.	Sales forecasting, risk assessment.
Decision Trees	Asks a series of true/false questions about features to sort data into outcomes.	Loan approval, medical screening.
K-Nearest Neighbors	Classifies a new data point based on what its closest data neighbors look like.	Product recommendations, simple pattern matching.

> **Summary:** Classical machine learning proves that you don't always need billions of parameters to solve smart problems—often, clean data and clever statistics are enough!

5. Review Checklist

- What is "feature engineering" and why is it necessary in classical ML?
- Who coined the term machine learning in the 1950s?
- What is one major advantage of classical machine learning over deep learning?